

Lab Exercise I: Data Pre-processing Using Jupyter Notebook

1. Load a dataset into a Jupyter Notebook using the Pandas library and display its basic information.
2. Identify the number of rows, columns, data types, and missing values in the dataset.
3. Perform exploratory data inspection by displaying the first, last, and randomly selected records of the dataset.
4. Detect and handle missing values using appropriate techniques such as deletion, mean, median, mode, or interpolation.
5. Identify duplicate records in the dataset and remove them.
6. Detect outliers using statistical methods (IQR or Z-score) and visualize them using box plots.
7. Handle outliers using suitable preprocessing techniques.
8. Convert categorical variables into numerical form using:
 - Label Encoding
 - One-Hot Encoding
9. Apply feature scaling techniques:
 - Min-Max Normalization
 - Standardization (Z-score Scaling)
10. Perform feature selection by identifying relevant attributes using correlation analysis.
11. Generate and interpret a correlation matrix and heatmap for the dataset.
12. Create new features from existing attributes through feature engineering.
13. Convert data types wherever necessary (e.g., string to datetime, integer to float).
14. Handle inconsistent and noisy data by applying suitable cleaning techniques.
15. Split the dataset into training and testing sets using Scikit-learn.
16. Perform data balancing using techniques such as random oversampling or SMOTE (if applicable).
17. Visualize the distribution of numerical features before and after pre-processing.
18. Compare the dataset before and after preprocessing and summarize the changes made.
19. Save the cleaned and preprocessed dataset as a new CSV file.
20. Document each preprocessing step performed in the Jupyter Notebook with appropriate explanations and outputs. (GitHub Account)